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SECTOR AT A GLANCE 02

Groundwater in India

Water in India

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GROUNDWATER IN INDIA

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The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Groundwater, defined as water stored below ground in aquifers and other sub-surface features, is critical to India. ~40% (447 Billion Cubic Meters or BCM) of India's total utilisable water resource annually (1137 BCM) comes from groundwater.¹ Groundwater supports 62% of irrigation needs, 85% of rural water supply, and 50% of urban water supply in the country.²

Adopting effective groundwater management practices in India can lead to improved irrigation water efficiency, greater agricultural productivity, and fulfilment of future socio-economic needs.

However, groundwater faces overexploitation and pollution. India contributes to 25% of global groundwater extraction, thus making it the largest extractor of groundwater since the 1980s.³ Moreover, fluoride levels beyond permissible limits have been detected in isolated pockets across 370 districts.⁴

Given these opportunities and challenges, groundwater quality, groundwater extraction, groundwater recharge, and groundwater governance have been key policy focus areas for India. Several ongoing and recent central government initiatives, including but not limited to Atal Bhujal Yojana, National Water Policy, National Water Mission, and Pradhan Mantri Krishi Sinchayee Yojana focus on sustainable management of groundwater resources.⁵ These efforts, along with several state- and local-level policies and initiatives, such as the Jalyukt Shivar Abhiyan 1.0 & 2.0 in Maharashtra, Neeru Chettu in Andhra Pradesh, or the Mukhya Mantri Jal Swavlamban Abhiyan in Rajasthan, have sought to ensure that India achieves a secure water future through the sustainable management of groundwater resources.

Figure 1: Key groundwater statistics⁶



Source: CGWB. (2023). National Compilation on Dynamic Ground Water Resources of India.

Note: * "Over-exploitation" here refers to groundwater extraction exceeding the annually replenishable groundwater recharge. ** "Stage of GW extraction" corresponds to the area under the "safe" category.

1 NITI Aayog, Composite Water Management Index 2.0, 2019

2 CGWB, Dynamic Ground Water Resources of India 2023, 2023

3 Ministry of Jal Shakti, Water Talk, accessed December 2023

4 PIB, Contamination of Ground Water, 2022

5 Ministry of Jal Shakti, Atal Bhujal Yojana, 2020

6 CGWB, National Compilation on Dynamic Ground Water Resources of India, 2023

Key trends

This section dives deeper into India's groundwater management landscape, outlining statistics and trends on:

1. Groundwater quality
2. Groundwater extraction
3. Groundwater recharge
4. Groundwater governance

Groundwater quality

Groundwater quality refers to the chemical, physical, and biological characteristics of underground water, encompassing factors like pH levels, mineral content, and microbial presence.

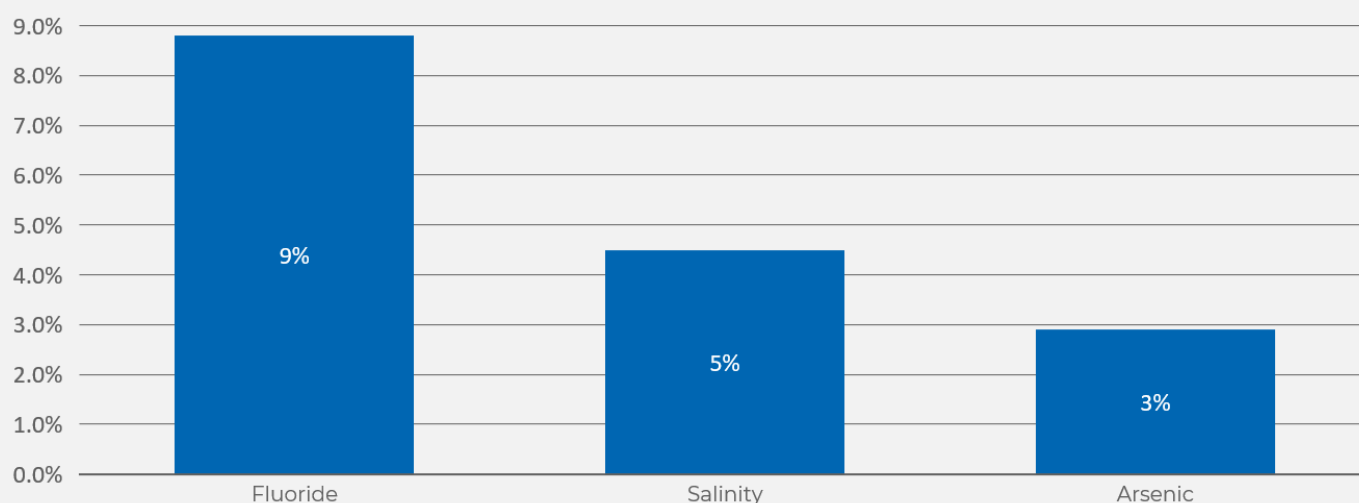
Contamination beyond permissible limits is present in one in ten assessment units:

Fluoride contamination was reported in 9% of assessment units, arsenic contamination in 4% and salinity in 7% of assessment units in 2023.⁷

The concentration of fluoride and arsenic contamination varies across states:

The states of Delhi, Gujarat, and Haryana have the highest percentage of assessment units affected by fluoride contamination. Assam and West Bengal have the highest percentage of assessment units affected by arsenic contamination.⁸

Figure 2: Share of assessment units affected by fluoride, salinity, and arsenic across states/UTs (2023)



Source: CGWB. (2023). Dynamic Ground Water Resources of India.

Groundwater extraction

Groundwater extraction encompasses the process of withdrawing water from underground sources, typically through wells or boreholes, to meet diverse needs such as agricultural irrigation, municipal supply, and industrial usage.

Extracted groundwater is used to meet agricultural, industrial, and domestic needs:

⁷ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

⁸ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

In 2022, 87% of the extracted groundwater was used for irrigation while the remaining 13% was for industrial and domestic uses.⁹

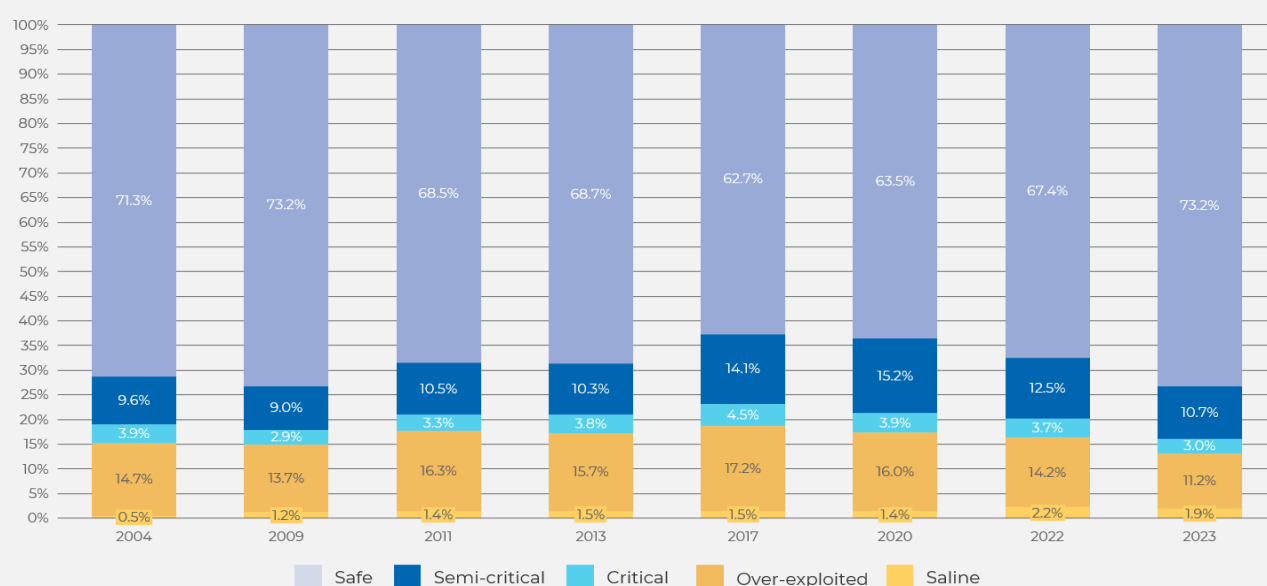
25% of assessed groundwater units in India are marked 'Semi-critical' or worse:

Out of the total 6553 groundwater units assessed in 2022, 11% have been categorized as 'Over-exploited' and 14% as 'Critical' and 'Semi-critical'.¹⁰

Extraction is more than extractable value in some states:

In states such as Haryana, Punjab, and Rajasthan, the stage of extraction of groundwater is over 100%, meaning the annual groundwater consumption is more than annual extractable groundwater resources.¹¹

Figure 3: Categorisation of assessment units based on status of groundwater quantity (2004-23)



Source: CGWB. (2023). Dynamic Ground Water Resources of India.

Groundwater recharge

Groundwater recharge involves the natural or artificial replenishment of underground aquifers through processes like infiltration from precipitation, irrigation, or deliberate practices to enhance water infiltration, aiming to maintain or restore groundwater levels.

Recharge rates in India have improved in recent years:

According to the Dynamic Groundwater Resource Assessment Report (2023), the annual groundwater recharge in 2023 was 449 BCM, an improvement from 438 BCM in 2022 and 432 BCM in 2017.¹²

Groundwater recharge is primarily driven by rainfall:

Rainfall is the major source of groundwater recharge, contributing 60% of the total annual recharge. The remaining 40% is contributed by other sources such as canal seepage, return flow from irrigation, and recharge from ponds, tanks, and water conservation structures.¹³

⁹ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

¹⁰ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

¹¹ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

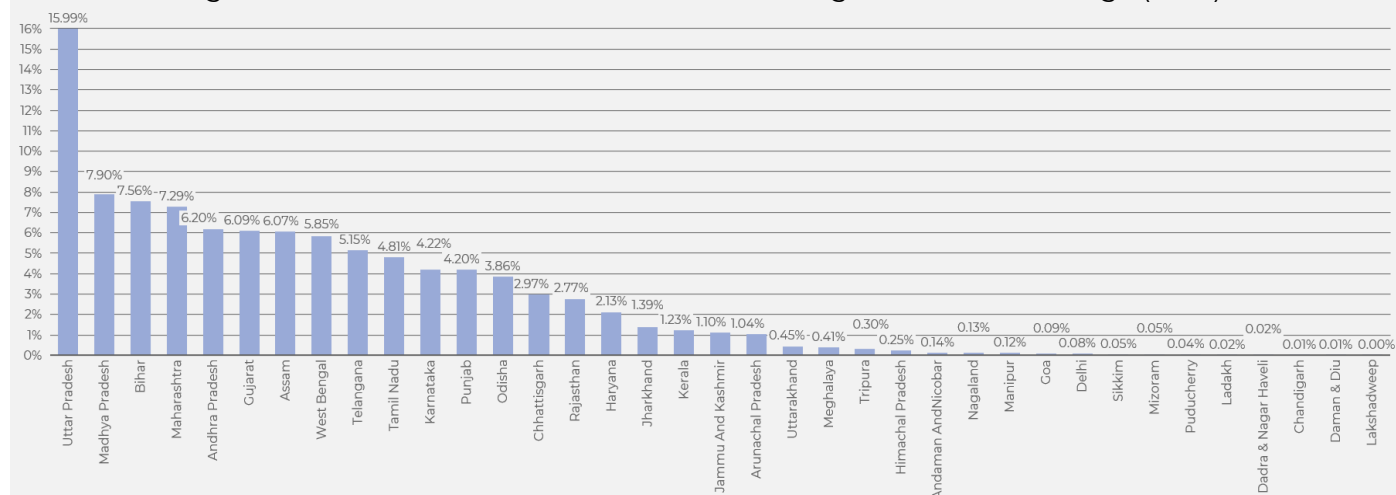
¹² CGWB. (2023). Dynamic Ground Water Resources of India 2023.

¹³ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

The depth of water level has improved for most analysed wells:

As per the Central Ground Water Board, a comparison of water level depth from pre-monsoon 2022 and 2012 indicates that ~67% of analysed wells show a rise in water level. The remaining show a decline in water levels. Groundwater level data of pre-monsoon 2022 reveals the general depth ranges from 5-10 m bgl.¹⁴

Figure 4: Contribution of states to India's annual groundwater recharge (2023)



Source: CGWB. (2023). Dynamic Ground Water Resources of India.

Groundwater governance

India's legal framework for groundwater governance and management is complex and multi-layered, with constitutional and statutory provisions of both the central and state governments and multiple departments under different ministries tasked with the execution of different components of water management.

Groundwater management is primarily state-driven:

Since water is a state subject, the management of groundwater resources is primarily the responsibility of India's states. State governments also work in tandem with the Centre, and state-level groundwater departments follow guidelines/policy frameworks provided by the central government. The Centre also supports states by circulating findings of various technical studies and pilot studies of CGWB for their consideration/implementation.¹⁵

The number of states adopting the Model Groundwater Bill has grown in recent years:

The Model Bill circulated in 2017 to enable appropriate state-level groundwater legislation for regulating its development and management has been adopted by 21 states/UTs, tailored as per their unique contexts.¹⁶

¹⁴ CGWB. (2023). Dynamic Ground Water Resources of India 2023.

¹⁵ Standing Committee on Water Resources. (2022-23). Groundwater: A Valuable but Diminishing Resource.

¹⁶ Ministry of Jal Shakti. (2023). Management of Ground Water.

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective groundwater management practices. These revolve primarily around the potential for improved irrigation water efficiency, greater agricultural productivity, and improved groundwater resilience.



Improvement in irrigation water efficiency:

Research studies suggest that optimal groundwater management for irrigation in India can lead to a 66% increase in the net benefit of irrigation. Optimal management refers to a scenario in which groundwater extraction decisions are made after considering the impact of such extraction as opposed to unregulated extraction. Replacing flat tariffs with metered cost pricing for groundwater pumping for agriculture can help achieve approximately half of this gain.¹⁷



Rise in agricultural productivity:

More efficient use of groundwater for irrigation with agroecological cropping systems can significantly increase the economic output of the agriculture sector. An analysis in Nature reveals that shifting to an agro-climatic zone suited to cropping can increase net farm incomes and arrest groundwater depletion while better meeting the country's food requirements, even without formal irrigation.¹⁸



Enhancement of groundwater resilience:

According to the Central Ground Water Board's 2022 assessment, there was an approximate 6 percentage point reduction in the number of overexploited groundwater units and a 10 percentage point increase in the number of safe category units compared to 2017. Such improvements in recharge are typically driven by greater recharge from canal seepage, water conservation structures including tanks and ponds, and return flow from irrigation. Some enabling efforts from the central government include the Jal Shakti Abhiyan, the Amrit Sarovar Mission, and the Atal Bhujal Yojana.¹⁹ The assessment also showed a reduction in annual extraction for irrigation, domestic & industrial uses, i.e. a declining trend of about 3.2% since 2017.²⁰ This can mean increased groundwater resilience and availability for future socio-economic needs.

Challenges

This section highlights some of the major challenges and constraints facing groundwater management in India. Efforts to resolve such challenges will ensure more efficient groundwater management, higher recharge rates, and greater internal coordination.



Inefficient waste management by industrial, agricultural, and other sectors:

90% of the industrial clusters in India report groundwater contamination due to the discharge of untreated effluents, especially from small-scale industries lacking equipment for wastewater treatment.²¹ Excessive fertilizers and pesticides used in agricultural activities also

¹⁷ Sayre, S. S., & Taraz, V. (2019). Groundwater depletion in India: Social losses from costly well deepening. *Journal of Environmental Economics and Management*, 93, 85-100.

¹⁸ Devineni, N., Perveen, S., & Lall, U. (2022). Solving groundwater depletion in India while achieving food security. *Nature Communications*, 13, 3374.

¹⁹ PIB. (2023). *Exploitation of Groundwater*.

²⁰ CGWB. (2023). *Dynamic Ground Water Resources of India*.

²¹ Lok Sabha Secretariat. (2023). *Standing Committee on Water Resources*.

lead to contamination of the drinking water supply. Additionally, improper solid and liquid waste management and disposal affect the quality of groundwater sources.²²



Over-reliance on water-intensive crops and inefficient irrigation practices:

87% of groundwater withdrawals in India are for irrigation, with extensive focus on crops like rice, sugarcane, and wheat- which alone account for 80% of the irrigation water supplied.²³ Furthermore, only 19% of the net irrigated area in India is cultivated through efficient micro-irrigation systems, highlighting the need to adopt such technologies.²⁴



Limited groundwater recharge infrastructure and adverse climate change effects:

The percentage of mapped areas covered with groundwater recharge infrastructure to the total mapped area was low in 2017-18.²⁵ Furthermore, the expansion of city infrastructure has led to reduced potential surface areas available for groundwater recharge, and projects such as stormwater drainage projects are yet to speed up.²⁶ Adverse effects of climate change like shifts in hydrological cycles also further worsen recharge rates.



Limited coordination and robust legislation:

There is a lack of coordination and data sharing between states, departments, and other authorities.²⁷ Further, only 50% of the states and union territories in India have groundwater legislation, and from these, only 20% have implemented the legislation partially due to a lack of clear guidelines from the Central ministry.²⁸

²² Lok Sabha Secretariat. (2023). Standing Committee on Water Resources; India Water Portal. (2021). Water contamination and pollution - A growing challenge for health and biodiversity.

²³ [https://loksabhadocs.nic.in/lsscommittee/Water Resources/17 Water Resources 22.pdf](https://loksabhadocs.nic.in/lsscommittee/Water%20Resources/17%20Water%20Resources%2022.pdf) Lok Sabha Secretariat. (2023). Standing

Committee on Water Resources; CGWB. (2023). Dynamic Ground Water Resources of India 2023.

²⁴ Ministry of Agriculture and Farmers Welfare. (2021). Agricultural Statistics at a Glance.

²⁵ NITI Aayog. (2019). Composite Water Management Index 2.0.

²⁶ Lok Sabha Secretariat. (2023). Standing Committee on Water Resources

²⁷ PRS. (2023). Standing Committee Report Summary - Groundwater: A Valuable but Diminishing Resource.

²⁸ Lok Sabha Secretariat. (2023). Standing Committee on Water Resources

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee groundwater management.



Ministry of Jal Shakti²⁹

The Ministry aims to effectively manage water as a national resource, which thus includes groundwater in its ambit. It functions through two departments- the first one being the Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) which is primarily responsible for establishing policy guidelines and programmes for the management, development, and conservation of water. The second department, i.e., the Department of Drinking Water and Sanitation provides technical and financial assistance to states to provide safe and adequate drinking water to rural India.

There are other important bodies under the ambit of the Ministry which are crucial to groundwater management. Primarily, these are:

- **Central Ground Water Board**- This is responsible for providing scientific and technical inputs for the management, exploration, monitoring, assessment, recharge, and regulation of groundwater resources of the country.
- **Central Groundwater Authority**- This aims to ensure the regulation, conservation, and protection of groundwater and to promote mass awareness.
- **National Institute of Hydrology**- This conducts research and development activities in water resources management (including groundwater) by collecting and analysing hydrological data, developing forecasting models, assessing water quality, etc.
- **National Water Informatic Centre**- This maintains a central repository of water resources in the country inclusive of both surface and groundwater resources.



Ministry of Agriculture & Farmers' Welfare

The Ministry aims to enhance irrigation infrastructure, promote efficient water use in agriculture, develop rainfed areas, support farmers in adopting sustainable water management practices for improved agricultural productivity, and formulate and implement policies and programs related to these areas.



Central Pollution Control Board

The Board is responsible for the prevention, control, and abatement of water pollution (thus including groundwater) in accordance with the Water (Prevention and Control of Pollution) Act, 1974.

Major policies and programs

This section highlights existing policies related to groundwater management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.

²⁹ Ministry of Jal Shakti. (Accessed December 2023). About the Ministry.



Atal Bhujal Yojana³⁰

The Atal Bhujal Yojana is the key government program in India aimed at the sustainable management of groundwater resources through community participation and water conservation activities. It entails the following key measures for sustainable groundwater management:

- Bringing community behaviour change through awareness initiatives and skill development to foster sustainable groundwater management
- Strengthening existing water management committees and converting them into Water User Associations that will conduct interventions such as water budgeting, community mobilisation, etc.



Pradhan Mantri Krishi Sinchayee Yojana, 2015³¹

PMKSY aims to promote the coordination of investments in irrigation at the individual field level through four sub-components: Accelerated Irrigation Benefit Programme, Har Khet Ko Pani, Per Drop More Crop, Watershed Development Component by:

- Expanding the cultivable land area with reliable access to irrigation
- Improving the efficiency of water usage on farms to minimise wastage and promoting the adoption of precision irrigation and other water-saving technologies
- Enhancing the replenishment of groundwater sources and implementing sustainable water conservation practices



National Water Mission

NWM aims at the conservation of water, minimisation of wastage, and more equitable distribution of water across and within States through integrated water resources development and management.³²

A noteworthy campaign launched by the NWM is the “**Jal Shakti Abhiyan: Catch the Rain**” aimed at saving and conserving rainwater by promoting rainwater harvesting structures. In 2023, the key theme, i.e., “Source sustainability for drinking water” envisions ensuring source sustainability in 150 water-stressed districts, as identified by the Jal Jeevan Mission.³³



Mission Amrit Sarovar³⁴

This is a recent initiative aimed at developing and rejuvenating 75 water bodies in each district of India. This was launched in 2022 as part of the ‘Azadi ka Amrit Mahotsav’ celebrations.



Ground Water Management and Regulation (GWMR) Scheme

This is a central sector scheme implemented by CGWB primarily including activities like aquifer mapping, groundwater level and quality monitoring, assessment of dynamic groundwater resources, regulation and control of groundwater withdrawal, and so on. **The National Aquifer Mapping and Management Programme (NAQUIM)** is one of the noteworthy activities under the scheme aimed at delineating and characterising the aquifers' geometry. It further looks at developing plans for sustainable groundwater management.

³⁰ Ministry of Jal Shakti, Government of India. (2020). Atal Bhujal Yojana.

³¹ Government of India. (2015). PMKSY.

³² Ministry of Jal Shakti, Government of India. (2011). National Water Mission.

³³ Ministry of Jal Shakti, Government of India. (Accessed December 2023). Jal Shakti Abhiyan: Catch the Rain.; Ministry of Jal Shakti, Government of India. (Accessed December 2023). Catch the Rain.

³⁴ Government of India. (Accessed December 2023). Mission Amrit Sarovar.

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. Managing groundwater resources sustainably is perhaps the single most important priority within the water sector in India. The solutions – better monitoring of water levels and quality, moving away from water-hungry cropping patterns, increasing the adoption of micro-irrigation systems, improved enforcement of water pollution regulation for industry, appropriate pricing of groundwater, appropriate pricing of agricultural electricity, community-based watershed management including the development of recharge infrastructure – are all important and are very widely known.

Several central government and state government schemes have made significant progress but continued policy action is needed to sustain these improvements. Some specific ideas that could add value include:

 Investing in community-level planning tools	Community-level scientific tools can help make better decisions related to the creation of water assets and also cropping patterns linked to water availability. A number of research institutions and NGOs have started developing tools that bring in hyperlocal natural resource data to enable this. State governments and other actors could actively seek out such partners and incorporate this within their groundwater management programs
 Increasing the density of water quality monitoring networks	The current system groundwater monitoring network, run by the CPCB, is invaluable but leaves out a large fraction of important water assets that need to be monitored; for instance, the current network has only 1235 groundwater stations or only one for approximately every 2,500 sq km of Indian territory or less than two for every district
 Scaling-up micro-irrigation programs	Comprehensive micro-irrigation programs, such as the Gujarat Green Revolution Company, can accelerate the adoption of micro-irrigation programs through a combination of financing, technical support, and ongoing maintenance.
 Investing in behaviour change programs for farmers and communities	Investing in behaviour change programs for farmers and communities in order to manage groundwater more effectively. Information and knowledge are sometimes not sufficient to drive behaviour and other approaches such as well-designed nudges can add significant value.



**Scaling up
the
monitoring of
industrial
water
extraction
and
discharge**

- Addressing industrial extraction of groundwater and effluent discharge will become critical especially as the scale of industrial activity grows and crowds out the water needs of the agriculture sector as well as human settlements.

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Recommended readings

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